

Open Airport Vector Grid (OAVG)

A Minimalist, Human-Readable Alternative to Coordinates & Plus Codes
System Specification — Version 2.0 (Draft) · September 2026

0. What Changed From v1

Version 2.0 keeps the core idea of v1 — an airport code as the origin, four clockwise sectors, positive distances, and precision set by code length — and closes the gaps found in review.

Gap in v1	v2 resolution	Section
Digits only reached ~10 km, but the zone is 50 miles (~80 km).	One extra digit per axis. Every precision level now covers 0–99,999 m.	4
No defined map projection, so two apps could produce different codes.	Local Transverse Mercator on WGS84, centred on the anchor. Fully specified.	3
No rule for points exactly on an axis.	Zero counts as positive (East / North).	3.3
No rounding rule; rounding can overflow the digit count.	Always truncate toward the anchor. Coarse codes are prefixes of fine codes.	4.3
Codes reset when crossing an axis (e.g. D→A).	Signed cell index rule for software arithmetic. Letters stay as-is for humans.	5
Overlapping zones gave one place two codes; remote areas had none.	Canonical anchor = nearest; alternates allowed; ICAO and Plus Code fallbacks.	6
“Airport terminal” origin; IATA codes can be reassigned.	Frozen, versioned OAVG Anchor Registry. Codes are never re-pointed.	2
“Under 50 KB” offline claim unqualified.	Footprint stated per registry scope.	9

1. Executive Summary

The **Open Airport Vector Grid (OAVG)** is a linear, variable-precision location code. It uses well-known **three-letter airport codes (IATA)** as the origin (0,0) of a local 2D grid. A location is written as the airport code, a sector letter showing the direction from the airport, and two distances in metres. A reader can tell at a glance roughly where a place is — e.g. “*about 3 km north-west of Trichy Airport*” — and software can convert codes to and from latitude/longitude offline.

TRZ-D03320331 = Trichy Airport, North-West, 3,320 m West, 3,310 m North (10 m cell)

2. Anchors & the Anchor Registry

- **Anchor:** an airport listed in the **OAVG Anchor Registry**. Primary anchors use the 3-letter IATA code of an airport with scheduled passenger service. Supplementary anchors use a 4-letter ICAO code (see Section 6). The code length tells a parser which type it is.
- **Anchor point:** the airport's published Aerodrome Reference Point (ARP), in WGS84 degrees to 7 decimal places, not the terminal building.
- **Frozen coordinates:** once registered, an anchor's coordinates never change, even if the airport is rebuilt or its official ARP moves. This keeps every code ever issued valid.
- **No re-pointing:** a registered code is permanently bound to its coordinates. Closed airports stay in the registry as *retired* (decode only). If IATA later gives the same code to a different airport, that airport is added under its ICAO code instead.

- **Registry format:** a versioned CSV — code, type, name, lat, lon, status, registered_on. Suggested seed source: OurAirports open data, cross-checked against each country's AIP.

TRZ, IATA, Tiruchirappalli International, 10.7651000, 78.7097000, active, 2026-09-28

Note: the TRZ coordinates above come from the v1 reference code. They must be checked against the ARP published in India's AIP before the registry is frozen, because every TRZ code depends on them.

3. Grid Geometry

3.1 Projection

Each anchor has its own flat grid, made with a **Transverse Mercator projection on the WGS84 ellipsoid**, centred on the anchor, with scale factor 1 and no false easting/northing. In PROJ notation:

```
+proj=tmerc +lat_0=<anchor lat> +lon_0=<anchor lon> +k=1 +x_0=0 +y_0=0 +ellps=WGS84 +units=m
```

This gives **E** (metres East, negative = West) and **N** (metres North, negative = South). Inside an 80 km zone, grid distance from the anchor matches true ground distance to within about 2 m. (The simple “degrees × 111,000” method used in v1 was off by 240–280 m at 80 km.)

3.2 Zone

- **Nominal zone:** 50 miles (80,467 m) from the anchor point.
- **Encodable range:** |E| and |N| each below 100,000 m. The extra room is used for alternates and gap-filling (Section 6).

3.3 Sectors (clockwise from North-East)

Sector	Direction	Rule	X digits mean	Y digits mean
A	North-East	$E \geq 0$ and $N \geq 0$	metres East	metres North
B	South-East	$E \geq 0$ and $N < 0$	metres East	metres South
C	South-West	$E < 0$ and $N < 0$	metres West	metres South
D	North-West	$E < 0$ and $N \geq 0$	metres West	metres North

Tie rule: a value of exactly zero counts as East or North. A point due north of the anchor is in A; the anchor itself is TRZ-A000000000.

4. Code Format & Precision

4.1 Structure

[ANCHOR]-[SECTOR][X][Y] X = East/West distance, Y = North/South distance

X and Y always have the **same number of digits** (d), with leading zeros kept. The cell size is $10^{(5-d)}$ metres.

4.2 Precision levels

Total digits	Per axis	Cell size	Use	Example (Trichy point, Sec. 7)
4	2 + 2	1 km	Area / locality	TRZ-D0303
6	3 + 3	100 m	Neighbourhood block	TRZ-D033033
8	4 + 4	10 m	Street / building (default)	TRZ-D03320331
10	5 + 5	1 m	Doorstep / pinpoint	TRZ-D0332403312

Every level covers the full 0–99,999 m range, so any point in the zone can be written at any precision.

4.3 Truncation (not rounding)

$X = \text{floor}(|E| \div \text{cell size})$, $Y = \text{floor}(|N| \div \text{cell size})$. A code names a **cell**, and a decoder returns the **centre** of that cell. Because digits are only ever cut off, a coarse code is always a prefix of the finer one:

TRZ-D0332403312 → TRZ-D03320331 → TRZ-D033033 → TRZ-D0303

To shorten a code, keep the first d digits of X and the first d digits of Y .

4.4 Canonical and display forms

- **Canonical:** upper case, no spaces — TRZ-D03320331. Parsers accept any letter case.
- **Display:** a dot may separate X and Y for easier reading — TRZ-D0332.0331. Both halves must be the same length.

- **Validation pattern:**

$^{\wedge}[A-Z]\{3,4\}-[ABCD](\backslash d\{2\}\backslash.?\backslash d\{2\}|\backslash d\{3\}\backslash.?\backslash d\{3\}|\backslash d\{4\}\backslash.?\backslash d\{4\}|\backslash d\{5\}\backslash.?\backslash d\{5\})\$$

5. Crossing Sectors

People read the sector letter; software never does arithmetic on it directly. To move, compare, or measure between codes, convert each to a **signed cell index**:

East/West: $i = X$ if sector is A or B; $i = -X - 1$ if C or D

North/South: $j = Y$ if sector is A or D; $j = -Y - 1$ if B or C

Add or subtract on (i, j) , then convert back with the same rule. The -1 makes the cell just west of the axis ($X = 0$ in D) sit next to the cell just east of it ($X = 0$ in A), so there is no gap or doubled cell at the axis.

Example at the 10 m level: one cell East of TRZ-D00000331 is TRZ-A00000331.

6. Overlaps & Gaps

- **Canonical code:** encoders use the **nearest active anchor** (by true ground distance to its anchor point). If two are exactly equally near, the alphabetically first code wins.
- **Alternate codes:** any active anchor whose encodable range (Section 3.2) contains the point may be used, e.g. a well-known city airport the reader recognises. Alternates are fully valid and decode to the same place. Apps should show the canonical code by default.
- **Beyond 80 km of any IATA anchor:** use the nearest anchor if it is within the 99,999 m range. Otherwise use the nearest supplementary ICAO anchor (4 letters, e.g. VOTR-...).
- **No anchor in range** (open ocean, polar regions): fall back to an Open Location Code (Plus Code), prefixed OLC:. This is the only case where OAVG does not produce its own code.

7. Worked Example: Trichy

An illustrative point near Trichy Central Bus Stand, at 10.7950461° N, 78.6793020° E, is 3,324.4 m West and 3,312.6 m North of the TRZ anchor. Its codes are in Section 4.2. At the default 10 m level (TRZ-D03320331), a 500 m move is 50 cells:

Move	Code	What changed
500 m North	TRZ-D03320381	Y + 50
500 m South	TRZ-D03320281	Y – 50
500 m East	TRZ-D02820331	X – 50 (moving toward the airport's north–south axis)
500 m West	TRZ-D03820331	X + 50 (moving away from the axis)
3.4 km East	TRZ-A00070331	Crosses the axis: sector D → A, X restarts from 0 (Section 5)

Note: the point is illustrative. Before publishing, replace it with the Bus Stand's surveyed coordinates and regenerate the codes with the reference implementation.

8. Conformance Test Vectors

A conforming implementation must reproduce these results with anchor TRZ = (10.7651000, 78.7097000).

Input	Expected output
encode(10.7950461, 78.6793020), 1 km	TRZ-D0303
encode(10.7950461, 78.6793020), 100 m	TRZ-D033033
encode(10.7950461, 78.6793020), 10 m	TRZ-D03320331
encode(10.7950461, 78.6793020), 1 m	TRZ-D0332403312
decode(TRZ-D03320331) [cell centre]	10.795068, 78.679296
decode(TRZ-D0332403312) [cell centre]	10.795045, 78.679301
encode grid E = 0, N = 1,500 m, 10 m	TRZ-A00000150 (tie rule)
encode grid E = -0.4 m, N = 1,500 m, 10 m	TRZ-D00000150
encode anchor point, 10 m	TRZ-A00000000
parse(trz-d0332.0331)	TRZ, D, X = 332, Y = 331, 10 m
move(TRZ-D03320331, +340 E cells)	TRZ-A00070331
any E or N ≥ 100,000 m	error: out of range for this anchor

Round-trip check: 20,000 random points across the full range, encoded at 1 m and decoded, all landed within 0.5 m of the original (half a cell, as expected).

9. Implementation Notes

- **Libraries:** Transverse Mercator is available everywhere — PROJ / pyproj (Python), proj4js (JavaScript), and standard Krüger-series formulas (the same maths as UTM) for dependency-free builds.
- **Offline footprint:** the encoder logic is a few KB. A compact binary registry takes about 12 bytes per anchor, so roughly 50 KB for commercial airports only and roughly 120–150 KB with supplementary ICAO anchors.
- **v1 compatibility:** v1 codes are **not** compatible. The same 4-, 6- and 8-digit lengths mean different cell sizes in v2, and the v1 document and v1 reference code disagreed with each other. v1 has no deployed user base, so v2 replaces it outright; decoders must not accept v1 codes.

10. Advantages Over Existing Systems

- **Human context:** unlike What3Words (*apple.chair.flying*) or Plus Codes (*PM5Q+CWQ*), an OAVG code tells a person the region and direction at a glance (“North-West of Trichy Airport”).
- **Simple maths:** once projected, moving, comparing and measuring are plain addition and subtraction on a local grid.
- **Offline:** no word dictionary and no cloud lookups; a small registry file is all that is needed.
- **Stable:** frozen anchors and truncation mean a code, once issued, always points to the same cell.

11. Open Items (Author Decisions)

- **Licensing alignment:** the reference code is currently under the Business Source License 1.1. That license makes the source available but is not an open-source license, which weakens v1's claim that no single party can restrict OAVG. One option is to publish the *specification* and *registry* under open licenses (e.g. CC BY 4.0 / CC0) while keeping the code on BSL. Get legal advice before deciding.
- **Check character:** consider an optional final check character to catch typing mistakes (e.g. swapped digits).
- **Vertical position:** building floor or altitude is out of scope for v2.
- **Registry governance:** who can add anchors, how the registry is versioned, and how disputes are handled.